Enterprise HR Analytics
& Employee Intelligence

End-to-End Data Analytics Portfolio Project

Analytics Workflow	Kaggle CSV → MySQL → Python → Excel → Power BI
Primary Focus	Workforce • Attrition • Performance • Employee Experience
Power BI Pages	HR Overview • Attrition & Workforce • Performance & Experience

Prepared as a portfolio-ready project report

1. Executive Summary

Enterprise HR Analytics & Employee Intelligence is an end-to-end HR analytics project designed to transform employee and performance data into actionable workforce insights. The project combines SQL analysis, Python-based data preparation and exploratory analysis, Excel reporting, and an interactive Power BI dashboard.

The analysis focuses on workforce size, employee attrition, compensation, tenure, employee satisfaction, performance ratings, training, and work-life balance. The final Power BI report provides interactive filtering across the main HR dimensions.

2. Dataset Overview

The project uses two related datasets: an employee master dataset and a performance-rating dataset.

Dataset	Rows	Columns	Purpose
Employee.csv	1,470	23	Employee demographics, job, compensation, tenure and attrition
PerformanceRating.csv	6,709	11	Employee satisfaction, training and performance ratings

EmployeeID is the key used to connect employee information with performance records.

3. End-to-End Analytics Workflow

Step	Stage	Main Activities
01	Kaggle Data	Source CSV datasets
02	MySQL	Database creation, loading, validation and SQL analysis
03	Python / Jupyter	Cleaning, validation, EDA and KPI preparation
04	Excel	Pivot analysis, KPI reporting and dashboard preparation
05	Power BI	Data modeling, DAX measures and interactive dashboard
06	Reports	Final portfolio documentation

4. Data Preparation & Quality Validation

Python was used to validate the source data before analysis. The datasets were checked for duplicates, invalid values, referential integrity, date types, whitespace issues, and logical tenure relationships.

- Exact duplicate employee records: 0
- Duplicate EmployeeIDs: 0
- Invalid Age, Salary and DistanceFromHome_KM values: 0
- Referential integrity between employees and performance records: 100%
- HireDate and ReviewDate converted to datetime
- Whitespace cleaned in selected categorical fields
- Clean datasets exported as employees_clean.csv and performance_clean.csv

Performance Data Coverage

The merged employee-level dataset contains 1,470 employees. Performance records are available for 1,280 unique employees, while 190 employees do not have performance records. The resulting 1,330 missing performance-related values are expected (7 performance fields × 190 employees) and were not treated as erroneous data.

5. Key HR KPIs

KPI	Value
Total Employees	1,470
Total Attrition	237
Attrition Rate	16.12%
Active Employees	1,233
Average Salary	112,956.50
Average Age	28.99
Average Years at Company	4.56
Average Performance Score	3.72
Average Job Satisfaction	3.42
Average Work-Life Balance	3.41
Average Self Rating	3.98
Average Manager Rating	3.47

6. Power BI Dashboard

The Power BI model connects employees_clean and performance_clean using EmployeeID. The relationship is many-to-one, active, with single-direction filtering from the employee table to performance records. DAX measures were created for the major HR KPIs.

Power BI Page	Purpose
HR Overview	Workforce and overall HR KPIs, including employee count, attrition, salary, performance and job satisfaction, with department-level visuals.
Attrition & Workforce	Attrition and workforce composition using KPIs for employees, attrition, attrition rate, active employees, tenure and age, plus department, gender, age, job-role and business-travel visuals.
Performance & Experience	Performance ratings, job satisfaction, work-life balance, training opportunities, job-role comparisons and manager-rating distribution.

Interactive Features

- Department, Gender and Attrition slicers on the HR Overview page.
- Department and Job Role filtering on analytical pages.
- Page navigation between all three Power BI pages.
- KPI cards and visuals update dynamically based on filter selections.

7. Business Insights

The project KPI calculations and dashboard analysis support the following observations:

- The workforce contains 1,470 employees, with 1,233 active employees and 237 employees recorded as attrition.

- The overall attrition rate is 16.12%, making retention a central area for workforce analysis.
- Average salary is 112,956.50, providing a baseline for compensation comparisons.
- Average tenure is 4.56 years, supporting analysis of employee experience and retention.
- Average self rating (3.98) is higher than average manager rating (3.47), creating a useful comparison point for performance discussions.
- Average job satisfaction is 3.42 and average work-life balance is 3.41, enabling employee-experience analysis alongside performance.
- Missing performance records represent unavailable data rather than automatically poor performance and should be handled accordingly.

8. Skills Demonstrated

Skill Area	Demonstrated Work
SQL / MySQL	Database creation, table design, data loading, validation and analysis
Python	Pandas-based cleaning, validation, merging, aggregation and EDA
Jupyter Notebook	Structured and reproducible analytical workflow
Excel	Pivot analysis, KPI reporting and dashboard preparation
Power BI	Data modeling, relationships, DAX, slicers, navigation and dashboard design
Data Analytics	KPI development, attrition analysis and business storytelling
GitHub	Portfolio project organization and version-controlled presentation

9. Project Structure

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Enterprise-HR-Analytics-Employee-Intelligence/
├── 01_Kaggle_Data/
├── 02_MySQL/
├── 03_Python_Jupyter/
├── 04_Excel/
├── 05_PowerBI/
├── 06_Reports/
├── 07_Visualizations/
├── 08_Icon/
└── README.md
```

10. Conclusion

This project demonstrates a complete data analytics workflow from raw HR datasets to an interactive business intelligence solution. The combination of MySQL, Python, Excel and Power BI shows the ability to prepare data, validate quality, analyze workforce patterns, calculate KPIs and communicate findings through an executive-friendly dashboard.

The project can be extended with time-based attrition trends, predictive attrition modeling, deeper compensation analysis, or automated refresh and reporting.

Prepared for portfolio presentation